Abstract Algebra

SF1678

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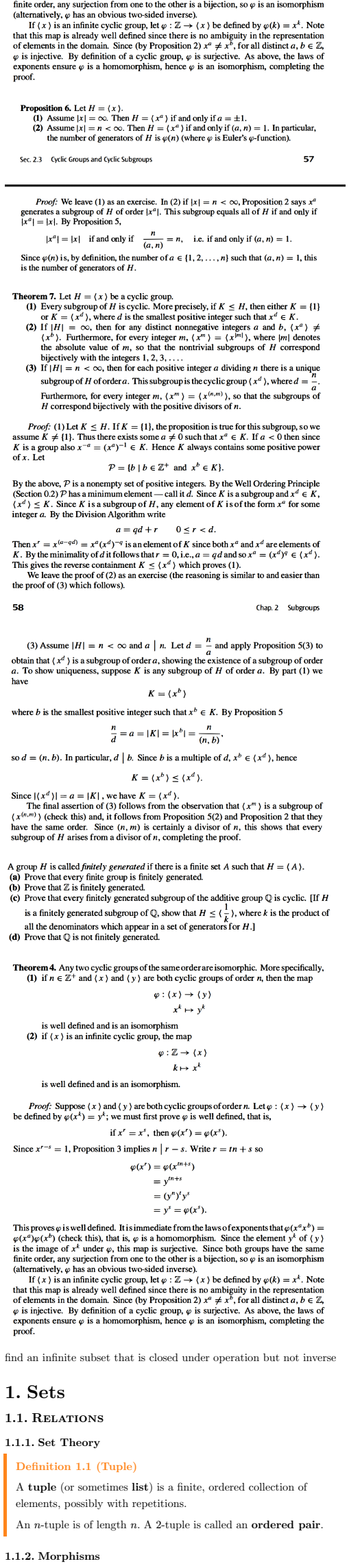
0.0.1. Excercises



Theorem 4. Any two cyclic groups of the same order are isomorphic. More specifically,
(1) if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map

$$\varphi : \langle x \rangle \to \langle y \rangle$$
$$x^k \mapsto y^k$$

is well defined and is an isomorphism
(2) if $\langle x \rangle$ is an infinite cyclic group, the map

$$\varphi : \mathbb{Z} \to \langle x \rangle$$
$$k \mapsto x^k$$

is well defined and is an isomorphism.

Proof: Suppose $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n. Let $\varphi : \langle x \rangle \to \langle y \rangle$ be defined by $\varphi(x^k) = y^k$; we must first prove φ is well defined, that is,

$$\text{if } x^r = x^s, \text{ then } \varphi(x^r) = \varphi(x^s).$$

Since $x^{r-s} = 1$, Proposition 3 implies $n \mid r - s$. Write $r = tn + s$ so

$$\begin{aligned}\varphi(x^r) &= \varphi(x^{tn+s})\\ &= y^{tn+s}\\ &= (y^n)^t y^s\\ &= y^s = \varphi(x^s).\end{aligned}$$

This proves φ is well defined. It is immediate from the laws of exponents that $\varphi(x^a x^b) = \varphi(x^a)\varphi(x^b)$ (check this), that is, φ is a homomorphism. Since the element y^k of $\langle y \rangle$ is the image of x^k under φ, this map is surjective. Since both groups have the same finite order, any surjection from one to the other is a bijection, so φ is an isomorphism (alternatively, φ has an obvious two-sided inverse).

If $\langle x \rangle$ is an infinite cyclic group, let $\varphi : \mathbb{Z} \to \langle x \rangle$ be defined by $\varphi(k) = x^k$. Note that this map is already well defined since there is no ambiguity in the representation of elements in the domain. Since (by Proposition 2) $x^a \neq x^b$, for all distinct $a, b \in \mathbb{Z}$, φ is injective. By definition of a cyclic group, φ is surjective. As above, the laws of exponents ensure φ is a homomorphism, hence φ is an isomorphism, completing the proof.

Proposition 6. Let $H = \langle x \rangle$.
(1) Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.
(2) Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of H is $\varphi(n)$ (where φ is Euler's φ-function).

Sec. 2.3 Cyclic Groups and Cyclic Subgroups 57

Proof: We leave (1) as an exercise. In (2) if $|x| = n < \infty$, Proposition 2 says x^a generates a subgroup of H of order $|x^a|$. This subgroup equals all of H if and only if $|x^a| = |x|$. By Proposition 5,

$$|x^a| = |x| \quad \text{if and only if} \quad \frac{n}{(a,n)} = n, \quad \text{i.e. if and only if } (a,n) = 1.$$

Since $\varphi(n)$ is, by definition, the number of $a \in \{1, 2, \ldots, n\}$ such that $(a, n) = 1$, this is the number of generators of H.

Theorem 7. Let $H = \langle x \rangle$ be a cyclic group.
(1) Every subgroup of H is cyclic. More precisely, if $K \le H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$.
(2) If $|H| = \infty$, then for any distinct nonnegative integers a and b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of H correspond bijectively with the integers $1, 2, 3, \ldots$.
(3) If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{(n,m)} \rangle$, so that the subgroups of H correspond bijectively with the positive divisors of n.

Proof: (1) Let $K \le H$. If $K = \{1\}$, the proposition is true for this subgroup, so we assume $K \neq \{1\}$. Thus there exists some $a \neq 0$ such that $x^a \in K$. If $a < 0$ then since K is a group also $x^{-a} = (x^a)^{-1} \in K$. Hence K always contains some positive power of x. Let

$$\mathcal{P} = \{b \mid b \in \mathbb{Z}^+ \text{ and } x^b \in K\}.$$

By the above, $\mathcal{P}$ is a nonempty set of positive integers. By the Well Ordering Principle (Section 0.2) $\mathcal{P}$ has a minimum element — call it d. Since K is a subgroup and $x^d \in K$, $\langle x^d \rangle \le K$. Since K is a subgroup of H, any element of K is of the form x^a for some integer a. By the Division Algorithm write

$$a = qd + r \qquad 0 \le r < d.$$

Then $x^r = x^{(a-qd)} = x^a (x^d)^{-q}$ is an element of K since both x^a and x^d are elements of K. By the minimality of d it follows that $r = 0$, i.e., $a = qd$ and so $x^a = (x^d)^q \in \langle x^d \rangle$. This gives the reverse containment $K \le \langle x^d \rangle$ which proves (1).

We leave the proof of (2) as an exercise (the reasoning is similar to and easier than the proof of (3) which follows).

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(3) Assume $|H| = n < \infty$ and $a \mid n$. Let $d = \frac{n}{a}$ and apply Proposition 5(3) to obtain that $\langle x^d \rangle$ is a subgroup of order a, showing the existence of a subgroup of order a. To show uniqueness, suppose K is any subgroup of H of order a. By part (1) we have

$$K = \langle x^b \rangle$$

where b is the smallest positive integer such that $x^b \in K$. By Proposition 5

$$\frac{n}{d} = a = |K| = |x^b| = \frac{n}{(n,b)},$$

so $d = (n, b)$. In particular, $d \mid b$. Since b is a multiple of d, $x^b \in \langle x^d \rangle$, hence

$$K = \langle x^b \rangle \le \langle x^d \rangle.$$

Since $|\langle x^d \rangle| = a = |K|$, we have $K = \langle x^d \rangle$.

The final assertion of (3) follows from the observation that $\langle x^m \rangle$ is a subgroup of $\langle x^{(n,m)} \rangle$ (check this) and, it follows from Proposition 5(2) and Proposition 2 that they have the same order. Since (n, m) is certainly a divisor of n, this shows that every subgroup of H arises from a divisor of n, completing the proof.

A group H is called *finitely generated* if there is a finite set A such that $H = \langle A \rangle$.
(a) Prove that every finite group is finitely generated.
(b) Prove that $\mathbb{Z}$ is finitely generated.
(c) Prove that every finitely generated subgroup of the additive group $\mathbb{Q}$ is cyclic. [If H is a finitely generated subgroup of $\mathbb{Q}$, show that $H \le \langle \frac{1}{k} \rangle$, where k is the product of all the denominators which appear in a set of generators for H.]
(d) Prove that $\mathbb{Q}$ is not finitely generated.

Theorem 4. Any two cyclic groups of the same order are isomorphic. More specifically,
(1) if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map $\varphi : \langle x \rangle \to \langle y \rangle$, $x^k \mapsto y^k$ is well defined and is an isomorphism
(2) if $\langle x \rangle$ is an infinite cyclic group, the map $\varphi : \mathbb{Z} \to \langle x \rangle$, $k \mapsto x^k$ is well defined and is an isomorphism.

Proof: Suppose $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n. Let $\varphi : \langle x \rangle \to \langle y \rangle$ be defined by $\varphi(x^k) = y^k$; we must first prove φ is well defined, that is, if $x^r = x^s$, then $\varphi(x^r) = \varphi(x^s)$. Since $x^{r-s} = 1$, Proposition 3 implies $n \mid r - s$. Write $r = tn + s$ so $\varphi(x^r) = \varphi(x^{tn+s}) = y^{tn+s} = (y^n)^t y^s = y^s = \varphi(x^s)$. This proves φ is well defined. It is immediate from the laws of exponents that $\varphi(x^a x^b) = \varphi(x^a)\varphi(x^b)$ (check this), that is, φ is a homomorphism. Since the element y^k of $\langle y \rangle$ is the image of x^k under φ, this map is surjective. Since both groups have the same finite order, any surjection from one to the other is a bijection, so φ is an isomorphism (alternatively, φ has an obvious two-sided inverse).

If $\langle x \rangle$ is an infinite cyclic group, let $\varphi : \mathbb{Z} \to \langle x \rangle$ be defined by $\varphi(k) = x^k$. Note that this map is already well defined since there is no ambiguity in the representation of elements in the domain. Since (by Proposition 2) $x^a \neq x^b$, for all distinct $a, b \in \mathbb{Z}$, φ is injective. By definition of a cyclic group, φ is surjective. As above, the laws of exponents ensure φ is a homomorphism, hence φ is an isomorphism, completing the proof.

find an infinite subset that is closed under operation but not inverse

1. Sets

1.1. RELATIONS

1.1.1. Set Theory

Definition 1.1 (Tuple)

A **tuple** (or sometimes **list**) is a finite, ordered collection of elements, possibly with repetitions.

An n-tuple is of length n. A 2-tuple is called an **ordered pair**.

1.1.2. Morphisms

Definition 1.2 (Map)

Subsets of the Cartesian product $A \times B$ are called **relations**. A **map** is a **well-defined** relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \to B \quad \text{or} \quad A \xrightarrow{f} B$$

and is

- **surjective** or **onto** if $f(A) = B$, which is equivalent to being **right invertible** with $f \circ f^{-1} = \mathrm{id}_B$.
- **injective** or **one-to-one** if $a_1 \neq a_2 \Longrightarrow f(a_1) \neq f(a_2)$, which is equivalent to being **left invertible** with $f^{-1} \circ f = \mathrm{id}_A$.
- **bijective** or a **one-to-one correspondence** if both above apply, which is equivalent to being **invertible**.

Definition 1.3 (Homomorphism)

A **homomorphism** is a map between algebraic structures that preserves the structure.

A group homomorphism $\varphi : G \to H$ satisfies

$$\varphi(g \star \gamma) = \varphi(g) \cdot \varphi(\gamma)$$

for all $g, \gamma \in G$.

Theorem 1.1

If $\varphi : G \to H$ is a group homomorphism,
(i) $\varphi(\mathrm{id}_G) = \mathrm{id}_H$
(ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G$
(iii) $\varphi(G) \le H$

Definition 1.4 (Isomorphism)

An **isomorphism** is a bijective homomorphism.

If there exists an isomorphism between two algebraic structures, they are said to be **isomorphic**, denoted with $\cong$.

Definition 1.5 (Endomorphism)

An **endomorphism** is a homomorphism from a mathematical object to itself.

Definition 1.6 (Automorphism)

An **automorphism** is simultaneously an isomorphism and an endomorphism: a bijective homomorphism from an object to itself.

Example. The **identity map** id_X is a trivial automorphism defined

$$\mathrm{id}_X(x) = x \qquad \forall x \in X.$$

Example. An automorphism of a set is called a **permutation**.

1.1.3. Equivalence

Definition 1.7 (Equivalence Relation)

An **equivalence relation** on a set A is a relation $R \subseteq A \times A$ with the following properties:
(i) **reflexive** $(a, a) \in R \quad \forall a \in A$
(ii) **symmetric** $(a, b) \in R \Longrightarrow (b, a) \in R$
(iii) **transitive** $(a, b), (b, c) \in R \Longrightarrow (a, c) \in R$
If $(x, y) \in R$ we write xRy or $x \sim y$.

Let R be an equivalence relation on a set A.

Definition 1.8 (Equivalence Class)

The **equivalence class** of an element $a \in A$

$$[a] = \bar{a} = \{b \in A \mid b \sim a\}$$

is the set of all elements related to it.

Example. The modulo m congruence class $[a] = \{a + km \mid k \in \mathbb{Z}\}$.

Definition 1.9 (Quotient Set)

The **quotient set** of A by R

$$A/R = \{[a] \mid a \in A\}$$

is the set of all equivalence classes induced by the relation.

Definition 1.10 (Partition)

A **partition** $\mathcal{P}$ of a set A is a collection of non-empty, disjoint subsets of A whose union is A.

Theorem 1.2

The quotient set A/R forms a partition of A.

Definition 1.11 (Quotient Map)

A **quotient map** is a map $\pi : A \to A/R$ defined by

$$\pi(a) = [a].$$

Theorem 1.3

Quotient maps are surjective.

1.2. $\mathbb{Z}$

1.2.1. Linear Diophantine Equations

Let a and b be integers with $\gcd(a, b) = d$.

Lemma (Bézout's Identity)

There exist integers x and y such that $ax + by = d$.

Lemma (Euclid's)

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Theorem 1.4 (Solvability Condition)

The linear Diophantine equation $ax + by = c$ has (infinitely many) solutions if and only if $d \mid c$.

Theorem 1.5 (Solutions)

The solutions, if they exist, are

$$x = x_0 + t\frac{b}{d}, \quad y = y_0 - t\frac{a}{d}, \quad \text{for } t \in \mathbb{Z}.$$

Theorem 1.6 (Euclidean Algorithm)

Let a and b be non-negative integers with $a \ge b$. Then there exist unique integers q and r such that

$$a = qb + r, \quad 0 \le r < b.$$

Example. We apply the algorithm to compute $\gcd(252, 105)$:

$$\begin{aligned}252 &= 2 \cdot 105 + 42\\ 105 &= 2 \cdot 42 + 21\\ 42 &= 2 \cdot 21 + 0\end{aligned}$$

Hence, $\gcd(252, 105) = 21$.

Example. We then use our results to solve $252x + 105y = 21$.

Since $\gcd(252, 105) = 21$, we now work backwards:

$$\begin{aligned}21 &= 105 - 2 \cdot 42\\ &= 105 - 2 \cdot (252 - 2 \cdot 105)\\ &= 105 - 2 \cdot 252 + 4 \cdot 105\\ &= 5 \cdot 105 - 2 \cdot 252\\ &= -2 \cdot 252 + 5 \cdot 105\end{aligned}$$

So a particular solution is $x_0 = -2, y_0 = 5$. The general solutions are

$$x = -2 + 5t, \quad y = 5 - 12t, \quad \text{for } t \in \mathbb{Z}.$$

2. Groups

2.1. DEFINITIONS

2.1.1. Groups

Definition 2.1 (Group)

A **group** $(G, \star)$ is a nonempty set G with a binary operation

$$\star : G \times G \to G$$

such that for all $a, b, c \in G$, it satisfies
(i) (Closure) $a \star b \in G$
(ii) (Associativity) $(a \star b) \star c = a \star (b \star c)$
(iii) (Identity) $\exists e \in G : e \star a = a \star e = a$
(iv) (Inverse) $\exists a^{-1} \in G : a \star a^{-1} = a^{-1} \star a = e$
(v) (Commutative) $a \star b = b \star a$
Additionally, a group is called **abelian** or **commutative** if it is

Examples. The **general linear group** $GL_n(\mathbb{R})$ consists of all matrices in $\mathbb{R}^{n \times n}$ with $\det \neq 0$. The **special linear group** $SL_n(\mathbb{R})$ additionally requires that $\det = 1$.

Tip

The proof of (iii) and (iv) can be shortened if (v) is proven.

Note

Defying conventional wisdom, **abelian** is usually not capitalized.

2.1.2. Subgroups

Let $(G, \star)$ be a group.

Definition 2.2 (Subgroup)

A subset $H \subseteq G$ is a **subgroup** if it is itself a group under the operation inherited from G.

Theorem 2.1 (Subgroup Criterion)

A subset $H \subseteq G$ is a subgroup if and only if
(i) $H \neq \varnothing$ (nonempty)
and
(ii) $a \star b^{-1} \in H \quad \forall a, b \in H$
or
(ii) $a \star b \in H \quad \forall a, b \in H$
(iii) $a^{-1} \in H \quad \forall a \in H$
or
(ii) $a \star b \in H \quad \forall a, b \in H$ (closed)
(iii) $|H| < \infty$ (finite)

Definition 2.3 (Maximal Subgroup)

A **maximal subgroup** H of a group G is a proper subgroup such that no proper subgroup K contains H strictly.

2.1.3. Cyclic

Definition 2.4 (Cyclic Subgroup)

For every $a \in G$, the set

$$\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$$

forms a **cyclic subgroup** of G.

Theorem 2.2

The cyclic subgroup $\langle a \rangle \le G$ is the smallest subgroup containing a.

Definition 2.5 (Generating Set)

The subgroup $\langle S \rangle = \langle s, s^{-1} \mid s \in S \rangle = \{s^k \mid s \in S, k \in \mathbb{N}\}$ consists of all *finite* products of elements in its **generating set** S and their inverses.

Definition 2.6 (Generator)

An element $a \in G$ is a **generator** of G if $\langle a \rangle = G$.

Definition 2.7 (Cyclic Group)

We say G is a **cyclic group** if it contains a **generator** of itself.

Theorem 2.3

All subgroups of a cyclic group are cyclic.

Theorem 2.4

Every cyclic group is abelian.

Proof. Let g and γ be elements of a cyclic group $G = \langle a \rangle$. Then there are integers m and n such that $g = a^m$ and $\gamma = a^n$, so

$$g\gamma = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = \gamma g.$$

2.2. $\mathbb{Z}/n\mathbb{Z}$

Theorem 2.5

If $ac \equiv bc \pmod{n}$ then $a \equiv b \pmod{n/\gcd(c, n)}$.

Let a and n be positive integers and let p be prime.

Definition 2.8 ($\mathbb{Z}/n\mathbb{Z}$)

The **integers modulo n** is a set of congruence classes

$$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/(n) = \mathbb{Z}_n = \{[0], [1], \ldots, [n-1]\}$$

that form an abelian group under $+$ and a monoid under $\times$, together forming a ring with a group of units called the **multiplicative group of integers modulo n**

$$U(n) = U(\mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})^\times = \{[a] \in \mathbb{Z}/n\mathbb{Z} \mid \gcd(a, n) = 1\}$$

with multiplicative inverses such that $aa^{-1} (\equiv a^{-1}a) \equiv 1 \pmod{n}$.

Definition 2.9 (Euler's Totient Function)

$$\phi(n) = |U(n)|$$

Example. $\phi(12) = |\{[1], [5], [7], [11]\}| = 4$

Theorem 2.6 (Euler's)

If a and n are coprime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof. By Lagrange's Theorem, the order $|[a]|$ of any element divides the order $|(\mathbb{Z}/n\mathbb{Z})^\times| = \phi(n)$ of its group, so $[a]^{\phi(n)} = [1]$.

Lemma

$\phi(p) = p - 1$

Theorem 2.7 (Fermat's Little)

If a and p are coprime, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

In general, even if they are not coprime,

$$a^p \equiv a \pmod{p}.$$

2.3. FAMILIES

2.3.1. Commutation

Let X be a nonempty subset of G.

Definition 2.10 (Centralizer)

The **centralizer** of X in G is the set

$$C_G(X) = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in X\},$$

all elements in G that commute with all elements in X, fixing every element of X under conjugation.

Remark

$gxg^{-1} = x \iff gx = xg$

Note

If X is a singleton set $\{x\}$, we simply write $C_G(x)$.

Definition 2.11 (Center)

The **center** of G is the set

$$Z(G) = C_G(G) = \{g \in G \mid g\gamma g^{-1} = \gamma \text{ for all } \gamma \in G\},$$

all elements in G that commute with all elements in G, fixing every element of G under conjugation.

Definition 2.12 (Normalizer)

The **normalizer** of X in G is the set

$$N_G(X) = \{g \in G \mid gXg^{-1} = X\},$$

all elements in G that commute with X as a set, fixing X as a set under conjugation.

Theorem 2.8

$Z(G) \le C_G(X) \le N_G(X) \le G$, with all equalities if G is abelian.

Theorem 2.9

If X is a subgroup, not just a subset, then $X \le N_G(X) \le G$.

Definition 2.13 (Commutator)

Let G be a group with elements $x, y \in G$ and nonempty subsets $A, B \subseteq G$.
(i) The **commutator** of x and y is $[x, y] = x^{-1}y^{-1}xy$.
(ii) $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$
(iii) Define $G' = \langle [x, y] \mid x, y \in G \rangle$, the subgroup of G generated by commutators of elements from G, called the **commutator subgroup** of G.

Theorem 2.10

$[x, y] = \mathrm{id}_G \iff x$ and y commute

2.3.2. Group Action

Definition 2.14 (Group Action)

A map $\cdot : G \times X \to X$ is a (**left**) **group action** if it satisfies:
(i) (Identity) $e \cdot x = x$ for all $x \in X$.
(ii) (Compatibility) $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

Let $\cdot : G \times X \to X$ be a group action.

Definition 2.15 (Orbit)

The **orbit** of $x \in X$ is the set

$$G \cdot x = \{g \cdot x \mid g \in G\},$$

all points in X that x can be moved to by the group action.

Definition 2.16 (Transitive)

The action of G on X is **transitive** if there is only one orbit, i.e., given any pair $x, y \in X$ there is some $g \in G$ such that $g \cdot x = y$.

Theorem 2.11

The orbits of a cyclic subgroup $\langle \sigma \rangle \le S_n$ is the sets of numbers in the distinct cycles given by its cycle decomposition.

Example. The orbits of $\langle (1\ 2)(3\ 4\ 5) \rangle$ are $\{1, 2\}$ and $\{3, 4, 5\}$.

Definition 2.17 (Stabilizer)

The **stabilizer** of $x \in G$ is the set

$$G_x = \{g \in G \mid g \cdot x = x\},$$

all elements in G that fix x under the group action.

The **stabilizer of** $X \subseteq G$ is the set

$$G_X = \{g \in G \mid g \cdot X = X\},$$

all elements in G that fix X under the group action.

Theorem 2.12 (Orbit–Stabilizer)

$|G \cdot x| = [G : G_x] = |G|/|G_x|$

Definition 2.18 (Fixed Point Set)

The **fixed point set** of $g \in G$ is the set

$$X^g = \{x \in X \mid g \cdot x = x\},$$

all elements in X that are fixed under g.

Definition 2.19 (Kernel)

The **kernel** is the set

$$\ker(\cdot) = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\},$$

all elements in G that fix every element of X under the group action.

Definition 2.20 (Faithful)

A group action is **faithful** if $\ker(\cdot) = \{e\}$.

2.4. CONJUGATION

2.4.1. Cosets

Let H be a subgroup of a group G.

Definition 2.21 (Coset)

For any $g \in G$, the **left coset** of H in G is

$$gH = \{g \star h \mid h \in H\}.$$

Similarly, the **right coset** is

$$Hg = \{h \star g \mid h \in H\}.$$

An element of a coset is a **representative** for that coset.

Theorem 2.13

The left and right cosets respectively partition G.

Definition 2.22 (Index)

The **index** of H in G is the number of left cosets, denoted $[G : H]$.

Theorem 2.14

There are equally many left and right cosets of H in G.

2.4.2. Conjugation

Definition 2.23 (Conjugation)

The **conjugation** by $a \in G$ is an **inner automorphism** of G defined $g \mapsto gag^{-1}$.

Definition 2.24 (Conjugate)

Two subsets X and Y of G are said to be **conjugate** if there is some $g \in G$ such that $X = gYg^{-1}$, i.e., if and only if they are in the same orbit of G acting on its subsets by conjugation.

Lemma

Let G be a group acting on a nonempty set A.
(i) The relation

$$a \sim b \quad \text{if and only if} \quad a = g \cdot b \text{ for some } g \in G$$

is an equivalence relation.
(ii) For each $a \in A$, the number of elements in its equivalence class (the orbit of G containing a) is $[G : G_a]$, the index of the stabilizer of a.
(iii) The stabilizer of any subset $S \subseteq G$ under conjugation is $G_S = N_G(S)$, the normalizer of S in G.

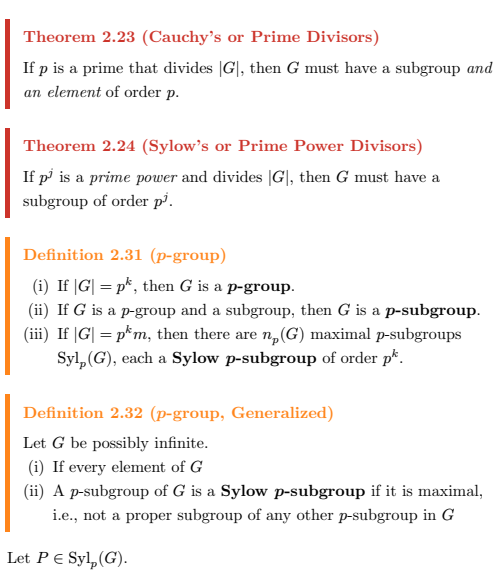
Lemma

$N_G(\{s\}) = C_G(\{s\}) = C_G(s)$ by definition.

Theorem 2.15

The number of conjugates of a subset S in a group G is $[G : N_G(S)]$, the index of the normalizer of S.

In particular, the number of conjugates of an element s of G is $[G : C_G(s)]$, the index of the centralizer of s.



Theorem 7. *(The Class Equation)* Let G be a finite group and let $g_1, g_2, \ldots, g_r$ be representatives of the distinct conjugacy classes of G not contained in the center $Z(G)$ of G. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|.$$

Proof: As noted in Example 2 above the element $\{x\}$ is a conjugacy class of size 1 if and only if $x \in Z(G)$, since then $gxg^{-1} = x$ for all $g \in G$. Let $Z(G) = \{1, z_2, \ldots, z_m\}$, let $\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_r$ be the conjugacy classes of G not contained in the center, and let g_i be a representative of $\mathcal{K}_i$ for each i. Then the full set of conjugacy classes of G is given by

$$\{1\}, \{z_2\}, \ldots, \{z_m\}, \mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_r.$$

Since these partition G we have

$$\begin{aligned}|G| &= \sum_{i=1}^{m} 1 + \sum_{i=1}^{r} |\mathcal{K}_i|\\ &= |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|,\end{aligned}$$

where $|\mathcal{K}_i|$ is given by Proposition 6. This proves the class equation.

Note in particular that all the summands on the right hand side of the class equation are divisors of the group order since they are indices of subgroups of G. This restricts their possible values (cf. Exercise 6, for example).

2.4.3. Normalness

Definition 2.25 (Normal Subgroup)

The following are equivalent for a subgroup H of G:
(i) H is a **normal subgroup** of G, denoted $H \trianglelefteq G$.

(ii) $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
(iii) $gHg^{-1} \subseteq H$ for all $g \in G$.
(iv) $gHg^{-1} = H$ for all $g \in G$.

(v) $gH = Hg$ (equal left coset and right coset) for all $g \in G$.
(vi) The sets of left and right cosets coincide.

Examples. Trivially, $G \trianglelefteq G$ and $\{e\} \trianglelefteq G$.

Warning

Normalness is not transitive, i.e.,

$$A \trianglelefteq B \trianglelefteq C \;\not\Rightarrow\; A \trianglelefteq C.$$

Note

Some authors use $\lhd$ exclusively for proper normal subgroups, while others use it for all normal subgroups.

The same principle holds for the subset sign $\subset$.

Theorem 2.16

Every subgroup of index 2 is normal.

Proof. Let $H \le G$ be the subgroup such that $[G : H] = 2$. We wish to show that $gH = Hg$ for all $g \in G$.

If $g \in H$, then $gH = H = Hg$. Otherwise, if $g \notin H$, then $gH = G/H = Hg$ since $H \cup Hg = G = H \cup gH$.

Thus, $gH = Hg$ for all g, so H is a normal subgroup of G.

Theorem 2.17

If $H \le G$ and $K \le G$, then $HK \le G$ if and only if $HK = KH$.

Corollary

If $gH = H\gamma$ for some $g, \gamma \in G$, then $g = \gamma$ and $g \in N_G(H)$.

Theorem 2.18

All subgroups of abelian groups are normal subgroups.

Definition 2.26 (Simple Group)

A group is **simple** if it has no normal nontrivial proper subgroup.

2.5. ORDER

2.5.1. Definition

Definition 2.27 (Order)

The **order of a group** $|G|$ is the number of elements in G.

The **order of an element** $|g|$ is the smallest positive integer n such that $g^n = e$. If there is no such n, the order is infinite.

Theorem 2.19 (Classification)

If G is a cyclic group of order n, then
(i) $G \cong \mathbb{Z} \iff n = \infty$
(ii) $G \cong \mathbb{Z}_n \iff n < \infty$

2.5.2. Lagrange's Theorem

Theorem 2.20 (Lagrange's)

Let G be finite. Then the order of H divides the order of G.

Corollary

$g^{|G|} = e$ for all $g \in G$.

Corollary

If G is of prime order, it
(i) lacks proper nontrivial subgroups
(ii) is generated by any element $g \neq e$
(iii) is cyclic

Theorem 2.21 (Lagrange's, Generalized)

Regardless of whether G is finite or infinite,

$$|G| = [G : H]\,|H|.$$

Proof. G has $|G|$ elements and is partitioned into $[G : H]$ distinct left cosets, each of which has $|H|$ elements.

Corollary

If $H \trianglelefteq G$ then $[G : H] = |G|/|H| = |G/H|$.

2.5.3. Torsion

Definition 2.28 (Bounded Group)

$\exists N \in \mathbb{Z}^+ : g^N = e \;\forall g \in G$

Definition 2.29 (Torsion Group)

$\forall g \in G \;\exists n \in \mathbb{Z}^+ : g^n = e$

Examples. Finite groups. Bounded groups.

Definition 2.30 (Torsion Subgroup)

If G is abelian, its torsion subgroup is an abelian subgroup

$$T(G) = \{g \in G : \exists n \in \mathbb{Z}^+ : g^n = e\} = \{g \in G : |g| < \infty\}.$$

Two special cases arise:
(i) If $T(G) = G$ then G is a **torsion abelian group**.
(ii) If $T(G) = \{e\}$ then G is a **torsion-free abelian group**.

2.5.4. Prime Divisors

Let G be a finite group, let p be a prime, and let k, j, and m be positive integers, where m (if it exists) is chosen such that $p \nmid m$.

Theorem 2.22

If p is the smallest prime that divides $|G|$, then any subgroup of index p (the existence of which is not guaranteed) is normal.

Theorem 2.23 (Cauchy's or Prime Divisors)

If p is a prime that divides $|G|$, then G must have a subgroup *and an element* of order p.

Theorem 2.24 (Sylow's or Prime Power Divisors)

If p^j is a *prime power* and divides $|G|$, then G must have a subgroup of order p^j.

Definition 2.31 (p-group)

(i) If $|G| = p^k$, then G is a **p-group**.
(ii) If G is a p-group and a subgroup, then G is a **p-subgroup**.
(iii) If $|G| = p^k m$, then there are $n_p(G)$ maximal p-subgroups $\mathrm{Syl}_p(G)$, each a **Sylow p-subgroup** of order p^k.

Definition 2.32 (p-group, Generalized)

Let G be possibly infinite.
(i) If every element of G
(ii) A p-subgroup of G is a **Sylow p-subgroup** if it is maximal, i.e., not a proper subgroup of any other p-subgroup in G

Let $P \in \mathrm{Syl}_p(G)$.

Lemma

If Q is a p-subgroup of G, then $Q \cap N_G(P) = Q \cap P$.

Theorem 2.25 (Sylow's)

(i) $\mathrm{Syl}_p(G) \neq \varnothing$.
(ii) If P is a Sylow p-subgroup and Q is a p-subgroup, then

$$\exists g \in G : Q \le gPg^{-1},$$

i.e., Q is contained in some conjugate of P. In particular, any two Sylow p-subgroups of G are conjugate in G.
(iii) $n_p \mid [G : P]$, where $[G : P] = m$.
(iv) $n_p \equiv 1 \pmod{p}$.
(v) $n_p = [G : N_G(P)]$ for every $P \in \mathrm{Syl}_p(G)$.

Corollary (Cauchy's Theorem)

If $p \mid |G|$ then there is a $g \in G$ such that $|g| = p$.

Cauchy's Theorem says G has a cyclical subgroup of order $|\langle g \rangle| = p$.

Corollary

Let $P \le G$ be a Sylow p-subgroup. The following are equivalent:
(i) P is the only Sylow p-subgroup of G, i.e., $n_p(G) = 1$.
(ii) P is normal in G.
(iii) P is characteristic in G.
(iv) All subgroups generated by elements of p-power order are p-groups, i.e., if X is any subset of G such that $|x|$ is a power of p for all $x \in X$, then $\langle X \rangle$ is a p-group.

Example. If $p \nmid |G|$, the only Sylow p-subgroup is the trivial group.

Example. If $|G| = p^k$, the only Sylow p-subgroup is G itself.

Example. If G is abelian, there is a unique Sylow p-subgroup for each p. This subgroup, the *p-primary component* of G, consists of all elements x whose order is a power of p.

Theorem 2.26

Sylow p-subgroups of the same group and order are isomorphic.

2.5.5. Finitely Generated Abelian Groups

Definition 2.33 (Finitely Generated)

Definition 2.34 (Free Abelian Group)

Theorem 2.27 (Fundamental of Finitely Generated Abelian Groups)

Let G be a finitely generated abelian group.
(i)

Definition 2.35 (Invariant Factor Decomposition)

Corollary

Theorem 2.28

Definition 2.36 (Elementary Divisor Decomposition)

Theorem 2.29

Let $m, n \in \mathbb{Z}^+$. Then $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1$.

Proof. We must prove both implications.

Consider the element $g = (1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ which has order $|g| = \mathrm{lcm}(|1_m|, |1_n|) = \mathrm{lcm}(m, n)$.

First assume $\gcd(m, n) = 1$. Then $|g| = mn$

Theorem 2.30

Tip

Definition 2.37 (Rank)

Definition 2.38 (Exponent)

2.6. HOMOMORPHISMS

2.6.1. Fibers

Definition 2.39 (Fiber)

Let $\varphi : G \to \Omega$ be a surjective map from a group to a set. The **fiber** of φ over some $\omega \in \Omega$ is its preimage in G

$$\varphi^{-1}(y) = \{g \in G \mid \phi(g) = \omega\},$$

the set of all elements in the domain that map to ω.

Let $\varphi : G \to H$ be a group homomorphism.

Definition 2.40 (Fiber of Group)

The **fiber** of φ over some $h \in H$ is

$$\varphi^{-1}(h) = \{g \in G \mid \varphi(g) = h\}.$$

Theorem 2.31

The collection of nonempty fibers forms a partition of G. Hence, every $g \in G$ is only in the fiber over $\varphi(g)$.

Definition 2.41 (Kernel)

The **kernel** of a group homorphism φ is the set

$$\ker \varphi = \varphi^{-1}(\mathrm{id}_H) = \{g \in G \mid \varphi(g) = \mathrm{id}_H\}.$$

Let $K = \ker \varphi$.

Theorem 2.32

$K \trianglelefteq G$

Theorem 2.33

For every $g \in G$ such that $\varphi(g) = h$ the entire fiber is

$$\varphi^{-1}(h) = gK = Kg.$$

In words, the fibers of a homomorphism are the cosets of its kernel.

Definition 2.42 (Quotient Group, Alternative)

The quotient group is

$$G/K = \{\varphi^{-1}(h) \mid h \in \varphi(G)\},$$

the set of all nonempty fibers of φ.

If φ is surjective, then $\varphi(G) = H$ and all fibers are nonempty.

Definition 2.43 (Quotient Group)

Let $N \trianglelefteq G$. The **quotient group** or **factor group** is

$$G/N = \{gN \mid g \in G\} (= \{Ng \mid g \in G\}),$$

the set of all left cosets of N in G, paired with the group operation

$$(aN) \star (bN) = (ab)N \quad \forall a, b \in G,$$

which is well-defined if and only if $N \trianglelefteq G$.

Example.

name	symbol	set	operation
large group	G	$\mathbb{Z}$	$+$
normal subgroup	N	$n\mathbb{Z}$	$+$
quotient group	G/N	$\mathbb{Z}_n$	$+ \bmod n$

2.6.2. Natural Projections

Definition 2.44 (Natural Projection)

Let $N \trianglelefteq G$. The natural/canonical projection/homomorphism is the map $\pi : G \to G/N$ defined by $\pi(g) = gN$ for all $g \in G$.

Theorem 2.34

For π defined above:
(i) π is a homomorphism
(ii) π is surjective, i.e., $\pi(G) = G/N$
(iii) $\ker \pi = N$

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[dr, "\Phi"'] & G/N \arrow[d, "\varphi"] \\ & H \end{tikzcd}$$

Definition 2.45 (Commute)

A diagram as above where all paths end at the same node is said to **commute**.

2.6.3. The Isomorphism Theorems

Let $\varphi : G \to H$ be a group homomorphism.

Theorem 2.35 (First or Fundamental of Homomorphisms)

(i) $\ker \varphi \trianglelefteq G$
(ii) $\varphi(G) \cong G/\ker \varphi$
(iii) $\varphi(G) \le H$

Corollary

(i) $|\varphi(G)| = [G : \ker \varphi]$

The following are equivalent:
(ii) φ is surjective
(iii) $H \cong G/\ker \varphi$
(iv) $|G| = [G : \ker \varphi]$ *Proof.* See (i).

The following are equivalent:
(v) φ is injective
(vi) $\ker \varphi = \mathrm{id}_G$

Lemma

If A and B are subgroups of G, then

$$AB \le G \iff AB = BA$$

Lemma

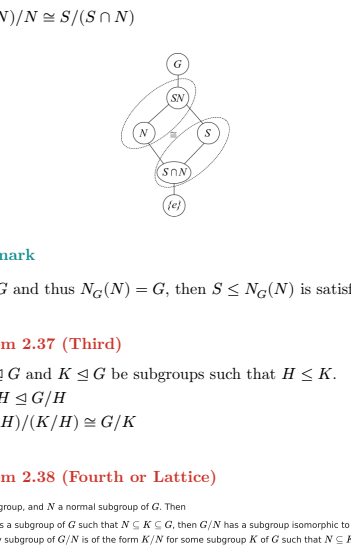
If A and B are finite subgroups of G, then

$$|AB| = \frac{|A||B|}{|A \cap B|}.$$

Note that AB is not necessarily a group.

Theorem 2.36 (Second or Diamond or Parallelogram)

Let $S \le G$ and $N \le G$ be subgroups such that $S \le N_G(N)$.
(i) $SN \le G$ *Proof.* See the above lemma.
(ii) $N \trianglelefteq SN$
(iii) $S \cap N \trianglelefteq S$
(iv) $(SN)/N \cong S/(S \cap N)$



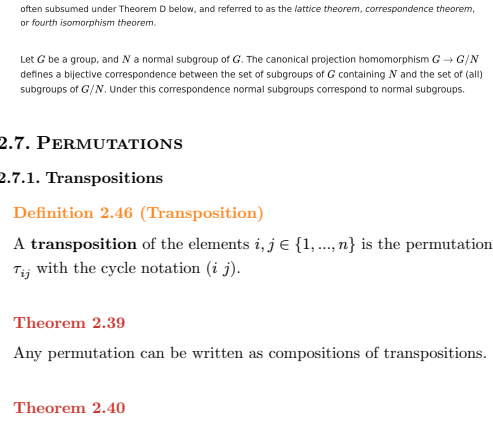
Remark

If $N \trianglelefteq G$ and thus $N_G(N) = G$, then $S \le N_G(N)$ is satisfied.

Theorem 2.37 (Third)

Let $H \trianglelefteq G$ and $K \trianglelefteq G$ be subgroups such that $H \le K$.
(i) $K/H \trianglelefteq G/H$
(ii) $(G/H)/(K/H) \cong G/K$

Theorem 2.38 (Fourth or Lattice)



Let G be a group, and N a normal subgroup of G. Then
1. If K is a subgroup of G such that $N \subseteq K \subseteq G$, then G/N has a subgroup isomorphic to K/N.
2. Every subgroup of G/N is of the form K/N for some subgroup K of G such that $N \subseteq K \subseteq G$.
3. If K is a normal subgroup of G such that $N \subseteq K \subseteq G$, then G/N has a normal subgroup isomorphic to K/N.
4. Every normal subgroup of G/N is of the form K/N for some normal subgroup K of G such that $N \subseteq K \subseteq G$.
5. If K is a normal subgroup of G such that $N \subseteq K \subseteq G$, then the quotient group $(G/N)/(K/N)$ is isomorphic to G/K.

The last statement is sometimes referred to as the *third isomorphism theorem*. The first four statements are often subsumed under Theorem D below, and referred to as the *lattice theorem, correspondence theorem*, or *fourth isomorphism theorem*.

Let G be a group, and N a normal subgroup of G. The canonical projection homomorphism $G \to G/N$ defines a bijective correspondence between the set of subgroups of G containing N and the set of (all) subgroups of G/N. Under this correspondence normal subgroups correspond to normal subgroups.

2.7. PERMUTATIONS

2.7.1. Transpositions

Definition 2.46 (Transposition)

A **transposition** of the elements $i, j \in \{1, \ldots, n\}$ is the permutation τ_{ij} with the cycle notation $(i\ j)$.

Theorem 2.39

Any permutation can be written as compositions of transpositions.

Theorem 2.40

The identity permutation may only be written as an even number of transpositions.

Theorem 2.41 (Order)

$|\sigma| = \mathrm{length}(\sigma)$

Definition 2.47 (Sign of a Permutation)

Let k be the number of transpositions in any decomposition of a $(k+1)$-cycle $\sigma \in S_n$. The **sign** of σ is $(-1)^k$.

Hence,
(i) if k is **even** then $\mathrm{sgn}(\sigma) = +1$ and σ is said to be even.
(ii) if k is **odd** then $\mathrm{sgn}(\sigma) = -1$ and σ is said to be odd.

Theorem 2.42 (Permutation Inverse)

Let $\sigma \in S_n$ be a permutation.
(i) $\sigma = (a_1\ a_2\ \cdots\ a_m) = \tau_1 \tau_2 \cdots \tau_k$ for some $m \le n$ and k
(ii) σ^{-1} exists

Proof. σ is a bijection.

(iii) $\sigma^{-1} = (a_m\ \cdots\ a_2\ a_1) = \tau_1 \tau_2 \cdots \tau_k$

Proof. Consider that $(\tau_i \tau_j)^{-1} = \tau_j^{-1} \tau_i^{-1} = \tau_j \tau_i$ and $(a\ b\ c)^{-1} = (a \to b \to c \to a)^{-1} = (a \leftarrow b \leftarrow c \leftarrow a)$.

(iv) $\sigma^{-1} = \sigma^k \iff \mathrm{length}(\sigma) = k$

Proof. The order of a k-cycle is k.

(v) $\mathrm{sgn}(\sigma^{-1}) = \mathrm{sgn}(\sigma) = (-1)^k$

Definition 2.48 (Fixed Point))

A **fixed point** is invariant under a transformation such as a permutation.

2.7.2. Symmetric Groups

Definition 2.49 (Symmetric Group)

The **symmetric group** of the finite set $X = \{1, 2, \ldots, n\}$ is

$$S_n = \{\sigma : X \to X \mid \sigma \text{ is bijective}\}$$

and consists of $n!$ permutations, which form **the symmetric group on n letters** under the operation of composition

Warning

Symmetric groups are not to be confused with *symmetry* groups.

Theorem 2.43

The automorphism group of a set is its symmetric group.

Definition 2.50 (Permutation Group)

A subgroup of a symmetric group S_n is a **permutation group**.

Definition 2.51 (Alternating Group)

The **alternating group**

$$A_n = \{\sigma \in S_n \mid \mathrm{sgn}(\sigma) = 1\}$$

is the subgroup of all ($n!/2$ if $n \ge 2$) even permutations in S_n.

Theorem 2.44

The number of k-cycles is

$$\binom{n}{k}(k-1)! = \frac{n!}{(n-k)!k!}(k-1)! = \frac{n!}{(n-k)!k}$$

in S_n. In A_n there are equally many if k is odd, but 0 if k is even.

Problem. Show that A_4 has no subgroup of order 6.

Solution. Assume the opposite; suppose there is a subgroup $H \le A_4$ such that $|H| = 6$.

By Lagrange's theorem, $[A_4 : H] = |A_4|/|H| = 12/6 = 2$. Theorem 2.16 dictates that any subgroup of index 2 is normal, so $H \lhd A_4$. Since the quotient group has order $|A_4/H| = [A_4 : H] = 2$,

$$(gH)^2 = H \implies g^2 H = H \implies g^2 \in H$$

for all $g \in A_4$.

There are 8 distinct 3-cycles in A_4. The inverse of any 3-cycle $g \in A_4$ is $g^{-1} = g^2 \in H$, since the length is 3. Hence, $g \in H$. However, there are 8 distinct 3-cycles in A_4, so it follows that $|H| \ge 8 > 6$.

Theorem 2.45

The sign map $\mathrm{sgn} : S_n \to \{1, -1\}$ is a group homomorphism.

The kernel of this homomorphism is the alternating group

$$\ker(\mathrm{sgn}) = A_n.$$

2.8. SYMMETRY

2.8.1. In Metric Spaces

Definition 2.52 (Isometry)

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is an **isometry** if it perserves distances, i.e.,

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)) \quad \forall x_1, x_2 \in X.$$

Definition 2.53 (Symmetry)

A **symmetry** of an object $O \subseteq X$ in a metric space (X, d) is an isometry $f : X \to X$ with the image $f(O) = O$.

Informally, a symmetry of a geometric object is a rearrangment of the object preserving the arrangement of its sides and vertices as well as its distances and angles.

Examples. Some types of symmetries:
(i) **Reflectional** Reflection across a hyperplane (a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$)
(ii) **Rotational** Rotation within a plane ($\mathbb{R}^2$ is rotated around a fixed point, while $\mathbb{R}^3$ is rotated around an axis).
(iii) **Translational** All points are shifted by a constant vector.
(iv) **Identity** No change.

Definition 2.54 (Rigid Motion)

A **rigid motion** is a map $f : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\|u - v\| = \|f(u) - f(v)\| \quad \forall u, v \in \mathbb{R}^n.$$

Equivalently, it is a surjective isometry between Euclidean spaces

Example. A symmetry of an object in an Euclidean space $X = \mathbb{R}^n$ is also a rigid motion.

2.8.2. Dihedral Groups

Definition 2.55 (Dihedral Group)

The **dihedral group** D_n or D_{2n} is the group of symmetries (or equivalently rigid motions) of an n-gon. Its degree is n and its order is $2n$.

Theorem 2.46

The dihedral group can be represented as

$$D_n = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

2.9. PRODUCTS

2.9.1. Direct

Definition 2.56 (External Direct Product)

The **(external) direct product** $G_1 \times G_2 \times \cdots \times G_n$ of the sets $G_1, G_2, \ldots, G_n$ with the operations $\star_1, \star_2, \ldots, \star_n$ is the set of n-tuples with operation defined componentwise:

$$(g_1, \ldots, g_n) \star (\gamma_1, \ldots, \gamma_n) = (g_1 \star_1 \gamma_1, \ldots, g_n \star_n \gamma_n).$$

If there are infinitely many groups, the n-tuples are replaced by sequences.

Theorem 2.47

$$G/G_i \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n$$

Theorem 2.48

$$|G_1 \times G_2 \times \cdots \times G_n| = |G_1|\,|G_2| \cdots |G_n|$$

Theorem 2.49

Let $x_i = (\mathrm{id}_1, \ldots, \mathrm{id}_{i-1}, g_i, \mathrm{id}_{i+1}, \ldots, n)$ for some $g_i \in G_i$. Then

$$|x_1 x_2 \cdots x_n| = \mathrm{lcm}(|x_1|, |x_2|, \ldots, |x_n|).$$

Let H and K be subgroups of a group.

Lemma

The number of distinct ways of writing each element of HK in the form hk is $|H \cap K|$

Proof.

Assume H and K are trivially disjoint and normal in the group, i.e.,

$$H, K \trianglelefteq G \quad \text{and} \quad H \cap K = \{e\},$$

Theorem 2.50 (Recognition)

$HK \cong H \times K$

Definition 2.57 (Internal Direct Product)

The internal direct product of H and K is HK (or, equivalently under the assumptions, KH).

3. Rings

3.1. DEFINITION

Definition 3.1 (Ring)

A **ring** is a set R with two binary operations $+$ and $\cdot$ such that
(i) $(R, +)$ is an abelian group with identity 0.
(ii) $(R, \cdot)$ is a monoid with identity 1.
(iii) Multiplication distributes over addition:
for all $a, b, c \in R$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

Definition 3.2 (Division Ring)

Definition 3.3 (Group of Units)

Let $(R, +, \cdot)$ be a ring. The **group of units** $U(R)$ is the set of all elements in R that have multiplicative inverses, i.e.,

$$U(R) = \{a \in R \mid \exists a^{-1} \in R : a \cdot a^{-1} = a^{-1} \cdot a = 1\}.$$

4. Fields

4.1. DEFINITION

Definition 4.1 (Field)

A **field** is a set F with two binary operations $+$ and $\cdot$ such that
(i) $(F, +)$ is an abelian group with identity 0.
(ii) $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1.
(iii) Multiplication distributes over addition:
for all $a, b, c \in F$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

Examples. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{R}(x), \mathbb{Q}(i)$, and $\mathbb{F}_p$ for prime p.

4.2. VECTOR SPACES

Definition 4.2 (Vector Space)

A vector space over a field F is a set V with two operations:

- Vector addition $+ : V \times V \to V$
- Scalar multiplication $\cdot : F \times V \to V$

These must satisfy
(i) (Closure under addition) $\mathbf{u} + \mathbf{v} \in V$
(ii) (Distributivity over vectors) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
(iii) (Associativity) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
(iv) (Identity) $\mathbf{u} + \mathbf{0} = \mathbf{u}$

(v) (Inverse) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
(vi) (Commutativity) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

(vii) (Closure under scalar multiplication) $a\mathbf{u} \in V$
(viii) (Distributivity over scalars) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
(ix) (Associativity) $a(b\mathbf{u}) = (ab)\mathbf{u}$
(x) (Identity) $1\mathbf{u} = \mathbf{u}$

Note that requiring closure is often omitted.